

SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2670–2681; original printed p. 89; physical source-PDF p. 79; recomposed running p. 71. The compiled reader is the output of the same corrected French edition, not an independent original typescript witness. Line 2682 is blank; the proof begins at line 2683 and is excluded.

Corollary 2.3. *Let X be a locally Noetherian prescheme that is locally embeddable in a regular prescheme, let Y be a closed subset of X , let $\mathcal{F}$ be a coherent $\mathcal{O}_X$ -module, and let n be an integer. The following conditions are equivalent:*

- (i) *for every $x \in U$, one has $\text{prof } \mathcal{F}_x > n - c(x)$;*
- (ii) *for every $x \in U$ such that $c(x) = 1$, one has $\text{prof } \mathcal{F}_x \geq n$;*
- (iii) *for every $i \in \mathbb{Z}$, $\mathcal{H}_Y^i(\mathcal{F})$ is coherent if $i \leq n$;*
- (iv) *$R^i i_*(\mathcal{F}|_U)$ is coherent for $i < n^{(4)}$.*

⁽⁴⁾*Editor's note.* This condition appeared only in the body of the proof, but not in the statement of the corollary; since it is used in Section 3, it has been added.